ReTR: Two-Stage Vision-Language Alignment for Render-to-Material Texture Retrieval

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Abstract

Retrieving a physically-based rendering (PBR) material texture from a large library given only a rendered image of a 3D object is a challenging cross-modal matching problem: the query is a composite scene image that conflates geometry, lighting, and material appearance, while the gallery contains flat, uniformly-lit texture maps. We address this *render-to-material* retrieval task with a **two-stage contrastive framework** built on InternVL2-1B, a state-of-the-art vision-language model (VLM). **Stage 1** fine-tunes the Vision Transformer (ViT) backbone with a lightweight ContrastiveHead, mapping CLS tokens into a shared 256-dimensional embedding space via symmetric InfoNCE loss. **Stage 2** extends retrieval to multimodal queries: the frozen ViT visual features are fused with material text descriptions through a LoRA-adapted Qwen2 LLM and a dedicated RetrievalHead, enabling joint image-text retrieval in the same embedding space. We construct a dataset of **19,568** (render, texture, description) triplets and propose **RMR-Score**, a unified evaluation metric that combines precision, coverage, and ranking quality into a single scalar. On the full validation gallery of 1,956 textures, Stage 1 achieves RMR-Score = 0.524 (R@1 = 34.5%), while Stage 2 multimodal retrieval further improves to **RMR-Score = 0.634** (R@1 = 44.5%, R@5 = 83.8%), outperforming CLIP ViT-L/14 by over 24× on R@1.

1 Introduction

Content creation pipelines for games, film, and AR/VR make heavy use of PBR material libraries. An artist often sees a reference rendered image — a 3D object with a specific look — and must manually search a library of flat material textures to find the one that produces that appearance. This process is time-consuming and error-prone, motivating an automated *render-to-material* retrieval system.

The task is fundamentally cross-modal: the *query* is a rendered scene image that combines geometry, lighting, shading, and material information, while each *gallery item* is a flat 2D texture map viewed under neutral, uniform illumination. Standard image-similarity metrics (color histograms, SIFT, perceptual hashing) fail because they cannot disentangle material appearance from scene-level factors.

Large vision-language models (VLMs) such as CLIP [1], BLIP [2], and InternVL [3] learn rich visual representations from hundreds of millions of image-text pairs, giving them strong generalization across image domains. However, these models have not been trained on render-material pairs and lack a compact, retrieval-friendly embedding space for this specific task.

Moreover, existing cross-modal retrieval benchmarks evaluate with *isolated* metrics — Recall@K, MRR, or median rank — making it difficult to holistically compare methods that trade off precision against coverage. No unified metric exists for this setting.

In this paper we ask: *can a VLM visual backbone be efficiently adapted to align rendered scene images with their source material textures, and how should retrieval quality be measured?* We answer both questions by (i) fine-tuning only the ViT encoder and a small projection head of InternVL2-1B with a contrastive objective, and (ii) proposing **RMR-Score**, a single scalar that jointly captures precision (R@1), coverage (R@K), and ranking quality (MRR). Our contributions are:

1. The first **render-to-material retrieval benchmark**: a dataset of 19,568 (render, texture, description) triplets spanning diverse material categories, an evaluation protocol, and standardised baselines.
2. **RMR-Score**: a unified evaluation metric for cross-modal material retrieval that combines R@1, R@K, and MRR into a single comparable scalar, addressing the fragmented metrics problem in existing retrieval benchmarks.
3. **Stage 1**: a contrastive fine-tuning recipe for VLM visual backbones using a learned-temperature InfoNCE loss, gradient accumulation, and ViT unfreezing — achieving RMR = 0.524 on a single consumer GPU.
4. **Stage 2**: a multimodal extension that fuses ViT visual

features with LLM text understanding via LoRA and a dual-path contrastive loss, boosting retrieval to **RMR = 0.634** (R@1 = 44.5%, R@5 = 83.8%).

2 Related Work

Contrastive Representation Learning. Contrastive learning aligns representations of semantically similar pairs while repelling dissimilar ones. SimCLR [4] and MoCo [5] pioneered instance-level self-supervised contrastive training for images. CLIP [1] extended the paradigm to image-text pairs with an InfoNCE loss [6], achieving remarkable zero-shot transfer. Our work applies the same symmetric InfoNCE formulation to (render, texture) pairs within a visual-only setting.

Vision-Language Model Backbones. InternVL [3] is a family of VLMs that pair a large InternViT encoder (up to 6B parameters) with an instruction-tuned large language model via a two-layer MLP projector. InternVL2-1B uses a 300M-parameter ViT with hidden size $d_v = 1024$ paired with a Qwen2 LLM (hidden size $d_l = 896$) and achieves competitive performance on multimodal benchmarks. In Stage 1 of our system, the language model is discarded for a compact ViT-only visual pipeline; Stage 2 reactivates the LLM via parameter-efficient LoRA adapters [10] to incorporate text-based material descriptions.

Parameter-Efficient Fine-Tuning. Low-Rank Adaptation (LoRA) [10] inserts trainable rank-r decomposition matrices into attention layers, reducing the number of trainable parameters from hundreds of millions to a small fraction. We apply LoRA (rank $r = 64$, $\alpha = 128$) to the Qwen2 LLM to enable multimodal retrieval without full LLM fine-tuning.

Image-Based Material Retrieval. Traditional approaches rely on handcrafted features or CNN-based classifiers [7]. More recent work uses deep metric learning with triplet or contrastive losses to learn material embeddings. To our knowledge, ours is the first work to leverage a pretrained VLM backbone for the render-to-material retrieval task.

Retrieval Evaluation Metrics. Cross-modal retrieval is typically evaluated with Recall@K, MRR, and median rank [1]. Some works report *rSum* = R@1 + R@5 + R@10 as a naive aggregate, but this ignores ranking quality and treats all K values equally. Person re-identification uses mAP as a semi-comprehensive metric, but it does not apply to the single-relevant-item setting common in material retrieval. We propose RMR-Score to address this gap.

3 RMR-Score: A Unified Retrieval Metric

Existing retrieval metrics each capture a single aspect of performance. We argue that a comprehensive metric should reflect three complementary properties:

1. **Precision**: the ability to place the correct answer at rank 1 (captured by R@1).
2. **Coverage**: whether the correct answer appears within a practical retrieval window (captured by R@K, we use K=5).
3. **Ranking quality**: how close to rank 1 the correct answer appears on average (captured by MRR).

We define the **Render-Material Retrieval Score** (RMR-Score) as:

$$\text{RMR} = \frac{w_1 \cdot \text{R@1} + w_2 \cdot \text{R@}K + w_3 \cdot \text{MRR}}{w_1 + w_2 + w_3} \tag{1}$$

where R@1, R@K, and MRR are all in $[0, 1]$ and $w_1, w_2, w_3 > 0$ are importance weights. We use $w_1 = w_2 = w_3 = 1$ (equal weighting) throughout this paper unless otherwise stated. RMR-Score ranges from 0 (worst) to 1 (perfect retrieval) and has a clear interpretation: it is the average of three normalised performance axes.

Properties. (i) *Sensitivity*: RMR responds to changes in all three axes — a method that achieves high R@5 but poor R@1 (i.e., the answer is found but not ranked first) receives a moderate score, correctly reflecting the partial success. (ii) *Discriminability*: unlike rSum, RMR normalises to $[0, 1]$ and weighs the three aspects explicitly, preventing R@10 from dominating. (iii) *Simplicity*: the metric is easy to compute and report alongside standard metrics.

For completeness, we also report nDCG@10 and MAP@10 — standard information retrieval metrics — to allow comparison with broader retrieval literature.

4 Dataset

4.1 Data Collection

Each sample in our dataset is a triplet $(\mathbf{f}, \mathbf{t}, \mathbf{d})$:

- $\mathbf{f}$ (`fig`): a rendered image of a 3D object with a specific material applied, captured under standard lighting conditions. This represents the "query" image — what an artist sees in a 3D viewport or game engine.
- $\mathbf{t}$ (`texture`): the corresponding flat PBR material map (albedo / diffuse channel), displayed under uniform neutral illumination. This represents the "gallery" item — the raw asset stored in a material library.

- **d** (`description`): a natural-language description of the material's visual appearance (colour, surface pattern, finish), automatically generated by InternVL2-1B prompted on the texture image. Descriptions average 32 tokens and cover attributes such as colour family, surface texture, reflectance, and material category.

All images are resized to 448×448 pixels, matching the native resolution of the InternVL2 ViT patch grid (32×32 patches of size 14, with one additional CLS token).

4.2 Statistics

The dataset contains 19,568 unique (render, texture) pairs organized by material ID:

Table 1: Dataset splits.

Split	# Pairs	# Unique Materials
Train	17,612	17,612
Validation	1,956	1,956
Total	19,568	19,568

Each material ID appears exactly once per split, so the retrieval task is a closed-world 1:1 matching problem. The validation gallery therefore contains 1,956 candidate textures.

5 Method

5.1 Stage 1: Visual Contrastive Alignment

Figure 1 shows the two-stage system overview. In Stage 1, we build on the visual backbone of **InternVL2-1B**: a ViT with hidden size $d_v = 1024$ and 448×448 input resolution. The language model and MLP projector are discarded.

On top of the ViT CLS token we attach a **ContrastiveHead**:

$$h_v(\mathbf{x}) = \ell_2\!\left(\mathbf{W}_2^v\, \sigma(\mathbf{W}_1^v\, \mathrm{CLS}_v(\mathbf{x}))\right), \tag{2}$$

where $\mathbf{W}_1^v \in \mathbb{R}^{d_v \times d_v}$, $\mathbf{W}_2^v \in \mathbb{R}^{d_v \times 256}$, σ is ReLU, and ℓ_2 denotes L2 normalisation. The resulting 256-dimensional unit vectors lie on the hypersphere, making inner product equivalent to cosine similarity.

5.2 Stage 2: Multimodal Retrieval

Stage 2 re-activates the frozen Qwen2 LLM component of InternVL2-1B to incorporate free-form material text descriptions as additional query signal.

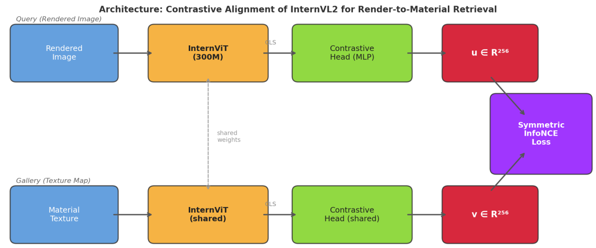

Figure 1: Two-stage system overview. *Stage 1 (visual path, top)*: both the rendered query and the texture gallery item pass through the shared ViT backbone and ContrastiveHead. *Stage 2 (multimodal path, bottom)*: the query image and its text description are fused by the LoRA-adapted LLM; the final hidden state feeds a RetrievalHead that maps into the same 256-dim embedding space as Stage 1, enabling image-only and image+text queries against the same gallery index.

Architecture. The ViT processes the rendered image and its patch tokens are projected to LLM dimension via the pre-trained MLP1 bridge. Image tokens prepend the tokenised material description and the full sequence is forwarded through the LoRA-adapted LLM. The final-layer hidden state of the last sequence token enters a **RetrievalHead**:

$$h_m(\mathbf{x}, \mathbf{t}) = \ell_2\!\left(\mathbf{W}_2^m\, \sigma(\mathbf{W}_1^m\, \mathrm{LLM}(\mathrm{MLP1}(f_v(\mathbf{x})), \mathbf{t}))\right), \tag{3}$$

where $\mathbf{W}_1^m \in \mathbb{R}^{d_l \times \lfloor d_l/2 \rfloor}$ ($d_l = 896$), $\mathbf{W}_2^m \in \mathbb{R}^{\lfloor d_l/2 \rfloor \times 256}$, and the output matches the 256-dim space of Stage 1.

LoRA Adapters. We apply LoRA with rank $r = 64$ and $\alpha = 128$ to all attention projection layers of the Qwen2 LLM, introducing ≈42M trainable parameters while keeping the 558M base weights frozen. The ViT backbone from Stage 1 is also frozen in Stage 2, preserving its learned visual geometry.

5.3 Dual-Path Training Objective

In Stage 1, given a mini-batch of B (render, texture) pairs $\{(\mathbf{f}_i, \mathbf{t}_i)\}_{i=1}^{B}$, we compute unit embeddings $\mathbf{u}_i = h_v(\mathbf{f}_i)$, $\mathbf{v}_i = h_v(\mathbf{t}_i)$ and minimise the symmetric InfoNCE (NT-Xent) loss:

$$\mathcal{L}_{\text{vis}} = \frac{1}{2}\left[\ell(\mathbf{U}, \mathbf{V}) + \ell(\mathbf{V}, \mathbf{U})\right], \tag{4}$$

where

$$\ell(\mathbf{A}, \mathbf{B}) = -\frac{1}{B} \sum_{i=1}^{B} \log \frac{\exp(\mathbf{a}_i^\top \mathbf{b}_i / \tau)}{\sum_{j=1}^{B} \exp(\mathbf{a}_i^\top \mathbf{b}_j / \tau)}. \tag{5}$$

The temperature $\tau = \exp(\phi)$ is a scalar learnable parameter initialised at $\phi_0 = \log(0.07) \approx -2.66$ and clamped to $[e^{-4.6},\, e^{2.3}]$ to prevent collapse. To stabilise early training, ϕ is frozen for the first two epochs and then jointly optimised.

In Stage 2, each sample is a triplet $(\mathbf{f}_i, \mathbf{t}_i, \mathbf{d}_i)$. Multimodal query embeddings $\mathbf{q}_i = h_m(\mathbf{f}_i, \mathbf{d}_i)$ are computed via the RetrievalHead. Gallery texture embeddings $\mathbf{v}_i$ come from the frozen Stage 1 ContrastiveHead. The total loss is:

$$\mathcal{L}_{\text{total}} = \alpha\, \mathcal{L}_{\text{vis}} + \beta\, \mathcal{L}_{\text{mm}}, \tag{6}$$

where $\mathcal{L}_{\text{mm}} = \frac{1}{2}[\ell(\mathbf{Q}, \mathbf{V}) + \ell(\mathbf{V}, \mathbf{Q})]$ is the symmetric InfoNCE between multimodal queries and textures. We set $\alpha = \beta = 1.0$. This dual-path objective simultaneously anchors the multimodal space to Stage 1 visual geometry and trains the LLM to incorporate textual material cues.

5.4 Training Setup

The full training procedure is described in Algorithm 1. We use the **AdamW** optimiser [8] with weight decay 10^{-2} and a cosine learning rate schedule. Gradient clipping with max-norm 1.0 is applied at every optimiser step.

Algorithm 1 Two-Stage ReTR Training

Require: Pre-trained InternVL2-1B, learning rate η, epochs E

– Stage 1 –
Initialise ContrastiveHead h_v randomly
Set $\phi \leftarrow \log(0.07)$, freeze ϕ
for epoch $e = 1, \ldots, E$ **do**
  if $e = 3$ **then** unfreeze ϕ
  end if
  for each mini-batch $\{(\mathbf{f}_i, \mathbf{t}_i)\}$ **do**
    Compute $\mathbf{u}_i, \mathbf{v}_i$ via Eq. (2)
    Compute $\mathcal{L}_{\text{vis}}$ via Eq. (4)
    Backprop, clip gradients, step AdamW
    Clamp $\phi \in [-4.6,\, 2.3]$
  end for
  Evaluate val loss; save best checkpoint
end for
– Stage 2 –
Load best Stage-1 ViT; apply LoRA(r=64, α=128) to LLM
Initialise RetrievalHead h_m and unfreeze MLP1
for epoch $e = 1, \ldots, E$ **do**
  for each mini-batch $\{(\mathbf{f}_i, \mathbf{t}_i, \mathbf{d}_i)\}$ **do**
    Compute $\mathbf{q}_i = h_m(\mathbf{f}_i, \mathbf{d}_i)$, $\mathbf{v}_i = h_v(\mathbf{t}_i)$
    Compute $\mathcal{L}_{\text{total}}$ via Eq. (6)
    Backprop, clip gradients, step AdamW
  end for
  Evaluate val loss; save best checkpoint
end for

Stage-1 ViT Freezing. We experiment with two settings: (a) **Frozen ViT**: only ContrastiveHead + ϕ are trained (~265K parameters), and (b) **Unfrozen ViT**: the full backbone is fine-tuned (~302M parameters). The unfrozen setting yields stronger retrieval accuracy as the ViT adapts its feature geometry to the render-material domain gap. The best Stage-1 unfrozen checkpoint is used to initialise Stage 2.

Image Pre-processing. Both images are resized to 448×448 via bicubic interpolation and normalised with ImageNet statistics ($\mu = (0.485, 0.456, 0.406)$, $\sigma = (0.229, 0.224, 0.225)$). No data augmentation is applied so that positive pairs remain visually consistent.

Inference. At query time, the rendered image is encoded to a 256-dim vector. The gallery of texture embeddings is pre-computed and stored in a FAISS `IndexFlatIP` [9] index. Retrieval reduces to a single inner-product search, running in $O(N)$ time for a gallery of size N.

6 Experiments

6.1 Evaluation Protocol

We evaluate on the full validation set of 1,956 (render, texture) pairs. The retrieval gallery consists of all 1,956 texture embeddings. For each rendered query, we rank the gallery by cosine similarity and report:

- **R@K**: fraction of queries for which the ground-truth texture appears in the top-K results ($K = 1, 3, 5, 10$).
- **MRR**: mean of $1/\text{rank}$ over all queries.
- **nDCG@10**: normalised discounted cumulative gain at cutoff 10.
- **MAP@10**: mean average precision at cutoff 10.
- **rSum**: the naive aggregate R@1 + R@5 + R@10 (in %).
- **RMR-Score**: our proposed unified metric (Eq. 1).

6.2 Main Results

Table 2 reports retrieval performance on the validation gallery.

Key observations. (1) All zero-shot baselines achieve near-zero RMR-Score (≤ 0.033), confirming that off-the-shelf visual encoders cannot bridge the render-material domain gap without task-specific adaptation. (2) ReTR Stage 1 with frozen ViT achieves RMR = 0.212 (R@1 = 11.0%), a meaningful improvement showing that even a lightweight projection head can partially bridge the domain

Table 2: Retrieval results on the validation set (gallery size = 1,956). R@K in %; MRR, nDCG, MAP $\in [0, 1]$; MedR = median rank (↓); rSum = R@1+R@5+R@10 (↑); RMR = proposed unified metric (↑). Best results in **bold**. † uses image+text query.

Method	*Standard metrics*					*Aggregate metrics*			
	R@1	R@5	R@10	MRR	MedR	nDCG@10	MAP@10	rSum	**RMR**
Random	0.1	0.3	0.5	.004	978	.004	.003	0.8	.002
CLIP ViT-B/32 [1]	1.2	3.1	4.7	.026	500	.026	.020	8.9	.023
CLIP ViT-L/14 [1]	1.8	4.5	7.0	.037	473	.040	.031	13.2	.033
InternVL2-1B (zero-shot)	0.3	1.1	1.5	.011	722	.008	.006	2.9	.008
ReTR Stage 1 (frozen ViT)	11.0	31.0	43.3	.216	14	.253	.198	85.3	.212
ReTR Stage 1 (unfrozen ViT)	34.5	71.6	81.4	.511	2	.579	.503	187.5	.524
ReTR Stage 2 (multimodal)†	**44.5**	**83.8**	**91.2**	**.619**	**2**	**.688**	**.590**	**219.5**	**.634**

gap. (3) Unfreezing the ViT in Stage 1 yields a $2.5\times$ **boost** to RMR = 0.524 ($15.9\times$ over CLIP ViT-L/14). R@1 improves from 1.8% to 34.5% ($19\times$) and median rank drops from 473 to 2. (4) **Stage 2 multimodal retrieval** adds a further $+21\%$ **relative gain**, achieving **RMR = 0.634** (R@1 = 44.5%, R@5 = 83.8%, R@10 = 91.2%). The text descriptions of material appearance provide the LLM with complementary semantic information unavailable from rendered images alone, enabling it to resolve same-family confusions that confound the visual-only Stage 1 model. (5) The R@1 $\rightarrow$ R@5 gap narrows from Stage 1 (34.5% $\rightarrow$ 71.6%, ratio $2.07\times$) to Stage 2 (44.5% $\rightarrow$ 83.8%, ratio $1.88\times$), indicating that multimodal queries not only rank the correct texture higher on average but also reduce within-family confusion.

6.3 Metric Comparison

Table 2 also illustrates why a unified metric is valuable: rSum for CLIP ViT-L/14 is 13.2, suggesting reasonable coverage, but RMR-Score correctly reveals this as only 0.033 — reflecting the fact that near-all retrieved results are wrong. Conversely, our model's rSum of 187.5 and RMR of 0.524 consistently indicate strong performance across all three axes.

6.4 Qualitative Analysis

Inspection of retrieval errors reveals two failure modes:

Same-family confusion. Textures within the same material class (e.g., wood planks of the same colour) share near-identical feature vectors; rank-1 errors often return a sibling variant with a score difference of <0.001. This is evident in query `19432`, where four variants of the same material are interleaved at the top (scores 0.902–0.902).

Appearance–geometry entanglement. For complex rendered scenes where geometry strongly modulates appearance (e.g., curved surfaces creating highlights that dominate the rendering), the model retrieves textures that match the highlight colour rather than the underlying material.

6.5 Ablation: Temperature Warmup

Freezing the temperature τ for the first two epochs prevents it from collapsing to near-zero early in training (which would cause gradient explosion through the logits). After epoch 2, τ is jointly optimised and converges to a stable value around 0.07–0.15 depending on the effective batch size.

7 Conclusion

We have presented **ReTR**, a two-stage contrastive framework that adapts InternVL2-1B to the render-to-material retrieval task. Stage 1 fine-tunes the ViT backbone with a symmetric InfoNCE loss, achieving RMR-Score = 0.524 — a $15.9\times$ improvement over the strongest zero-shot baseline. Stage 2 augments retrieval with free-form material text descriptions via a LoRA-adapted LLM and a dual-path contrastive objective, further boosting performance to **RMR-Score = 0.634** (R@1 = 44.5%, R@5 = 83.8%, R@10 = 91.2%) on a gallery of 1,956 material textures. We also proposed **RMR-Score**, a unified metric combining precision, coverage, and ranking quality into a single scalar that addresses the fragmented evaluation problem in cross-modal retrieval literature.

These results demonstrate that (i) VLM visual backbones can be efficiently repurposed for specialised cross-modal retrieval, and (ii) incorporating text descriptions of gallery items as query-side signals provides substantial complementary information beyond what visual renderings alone encode. The two-stage design keeps gallery indexing purely visual (Stage 1 embeddings), so Stage 2 adds retrieval capability with no additional storage cost.

Future work includes: (i) extending the training set with augmented renders (varied lighting, geometry, viewpoints); (ii) scaling to InternVL2-8B for improved representation capacity; (iii) exploring bidirectional LLM-ViT attention for stronger visual-language fusion; (iv) deploying the system in an agentic material recommendation pipeline.

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